

MANCHESTER
1824

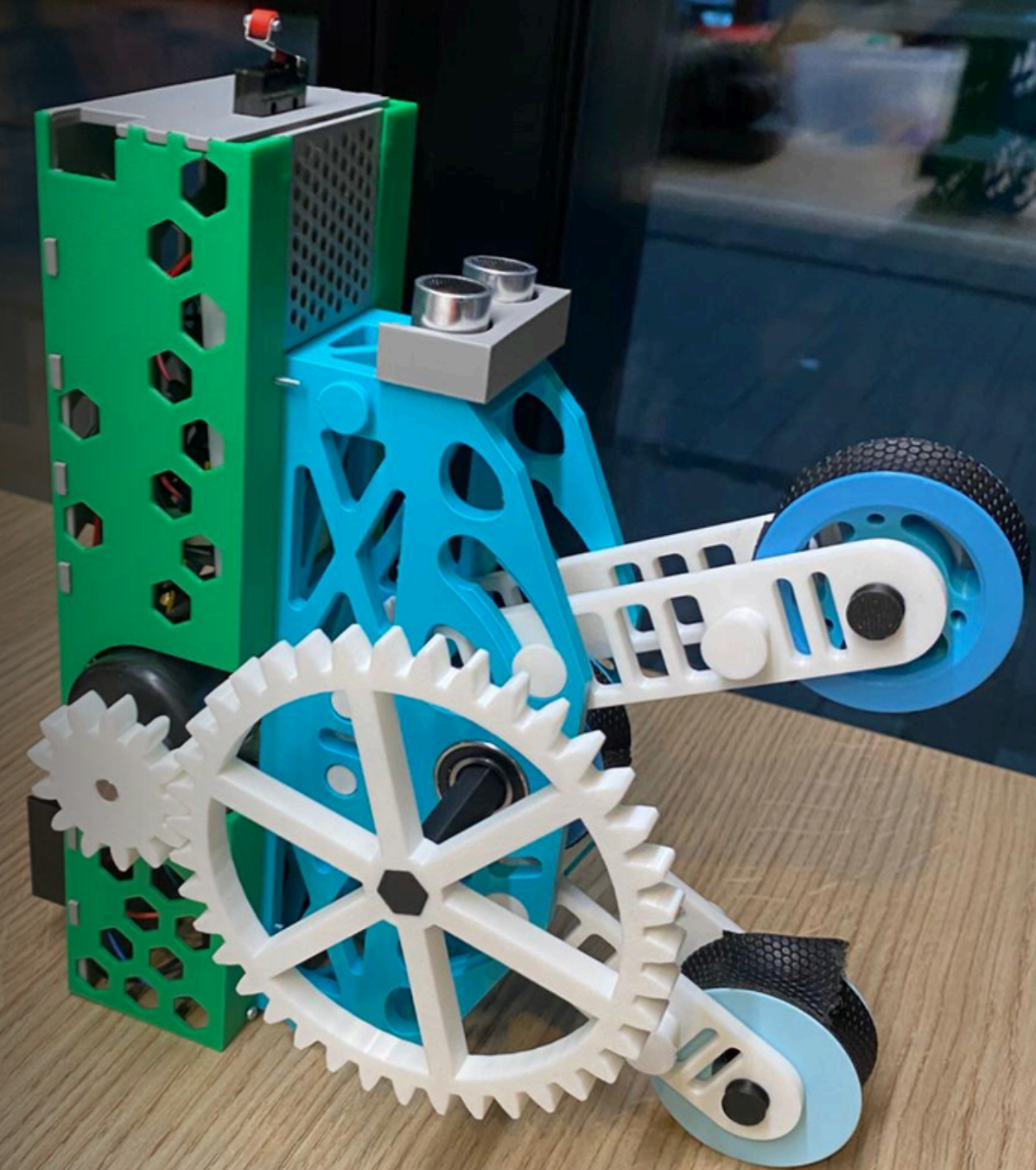
The University of Manchester

Institution of
**MECHANICAL
ENGINEERS**

IMECHE DESIGN CHALLENGE - ADVANCED CATEGORY

***THE UNIVERSITY OF MANCHESTER
UNLIMITED RICE PUDDING***

PRESENTED BY:
CHIEMEZIELAM SAMUEL AHIWE
KEI HANG DEREK CHAN
AHMED MISHQAT
ANTHONY YU

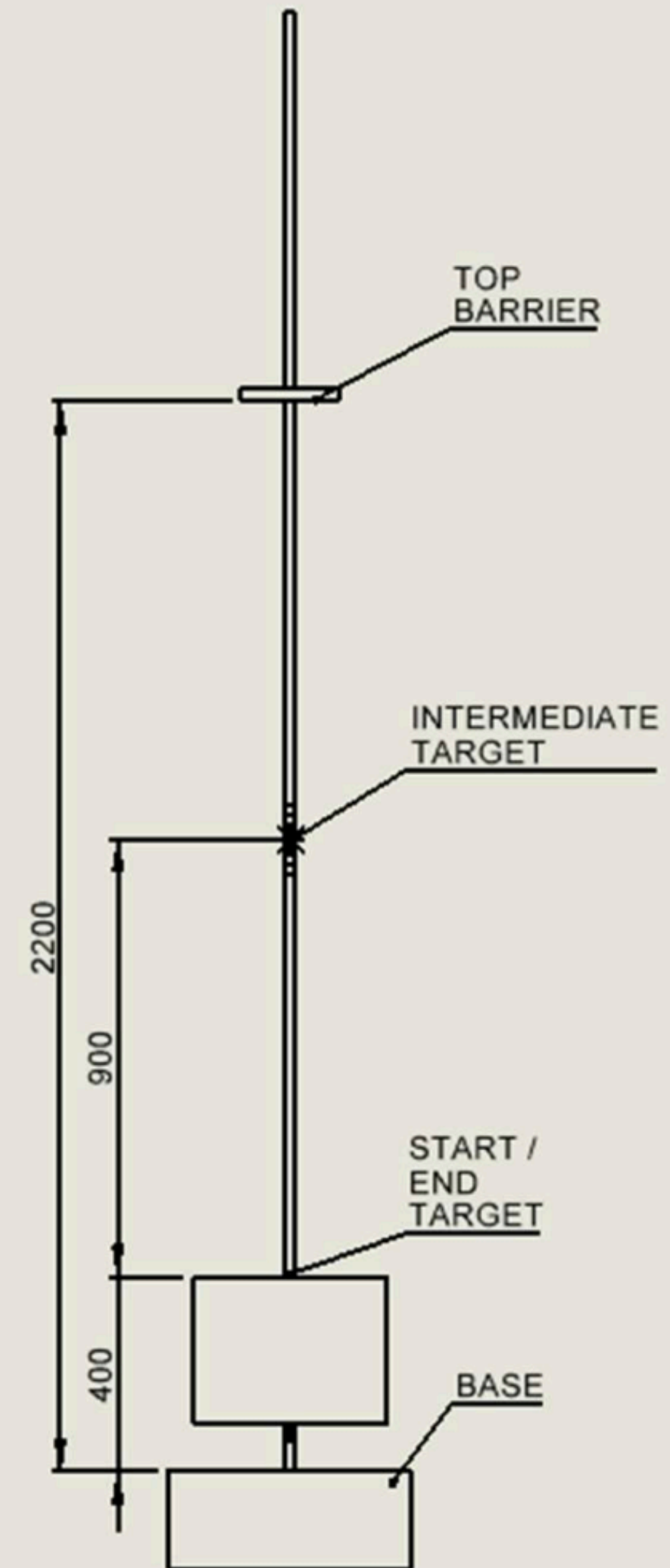


AIM - ADVANCED CHALLENGE

It lifts and lowers a **progressively changing load** in the form of a chain up and down the pipe, before releasing it and travelling back up, reaching as close as possible to the **intermediate target**.

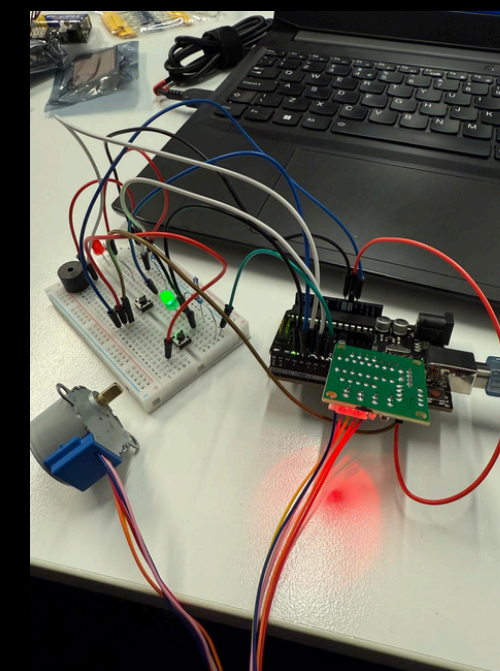
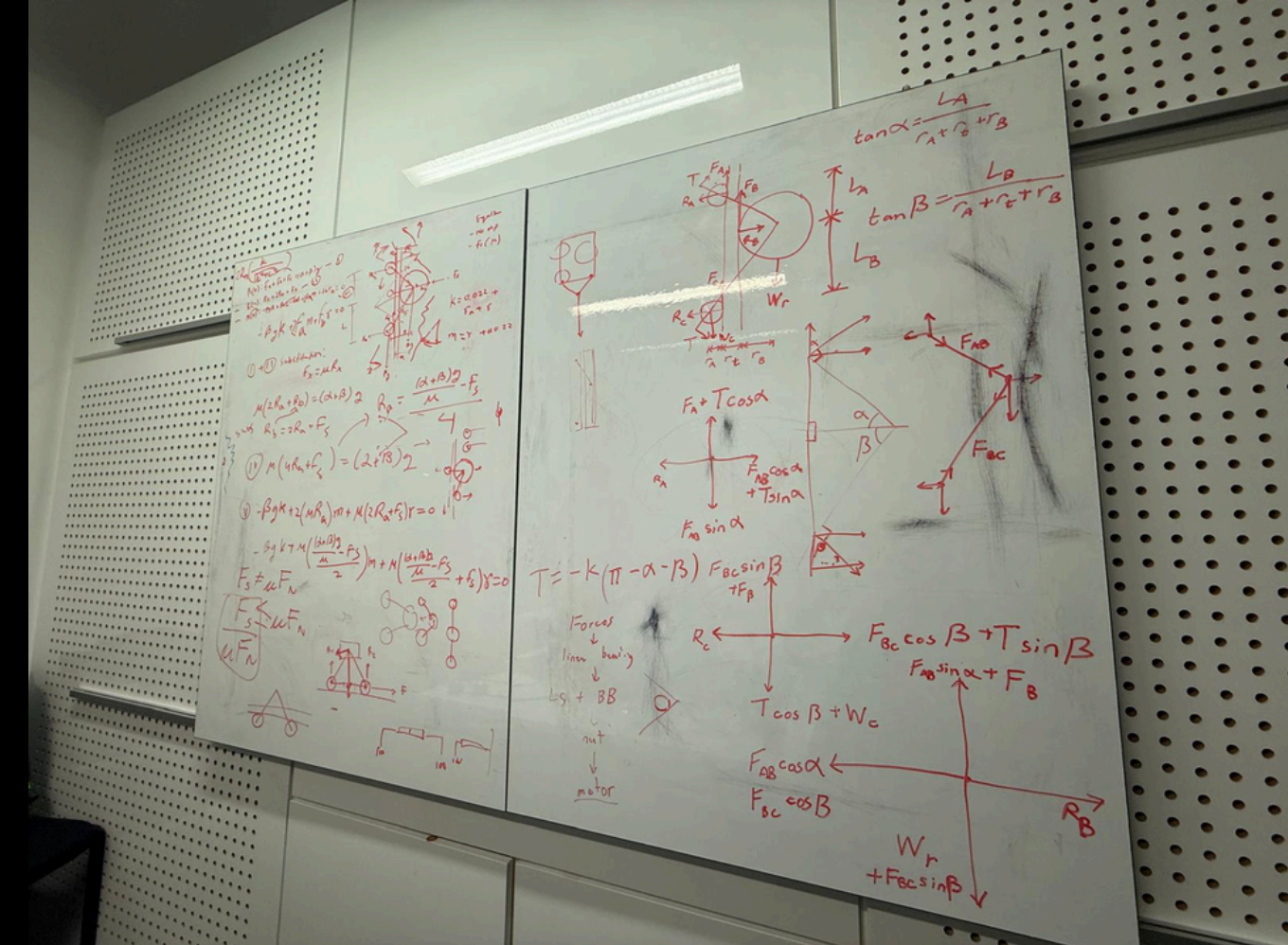
Main considerations:

1. Preventing slip
2. Accuracy
3. Cost and material usage
4. Size
5. Effective release



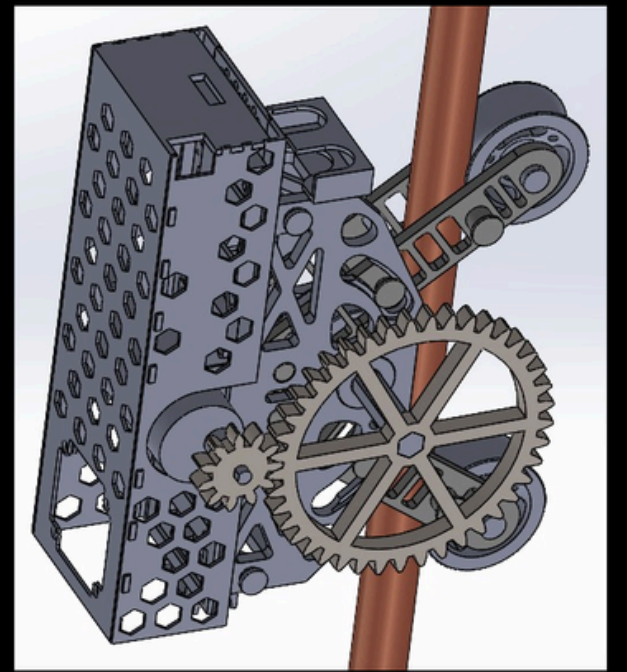
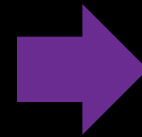
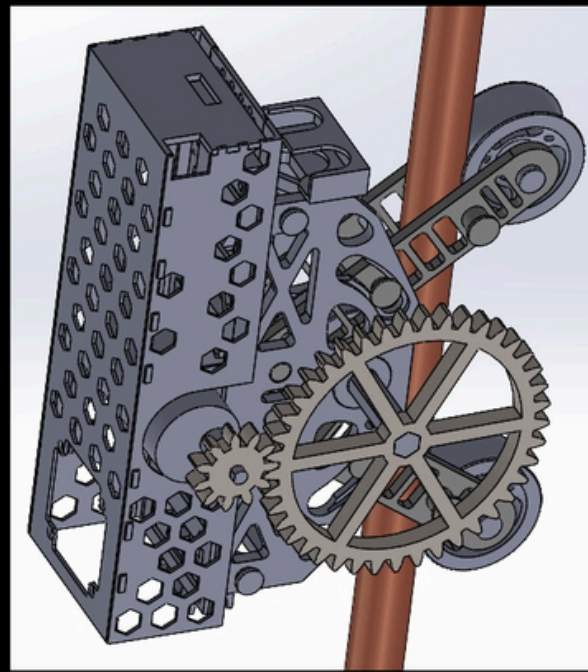
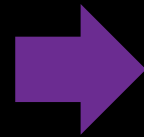
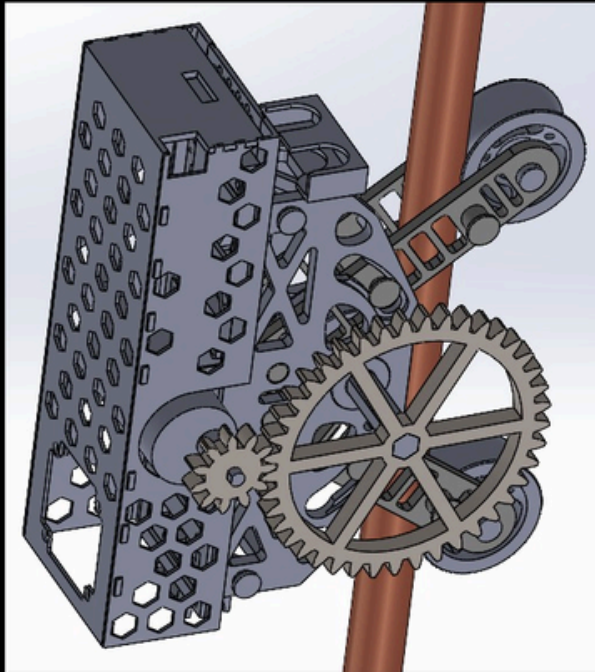
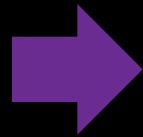
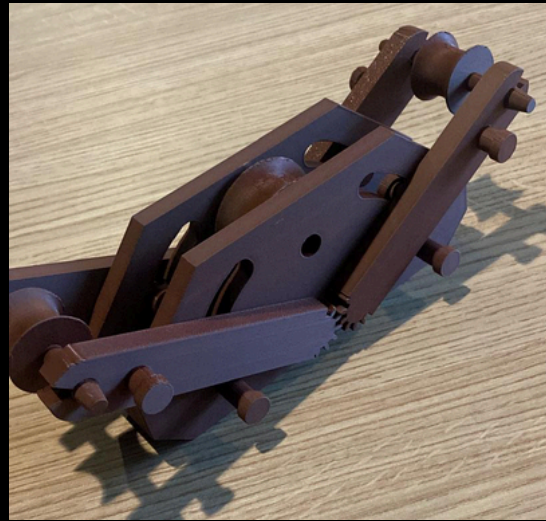
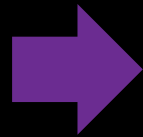
DESIGN APPROACH

1. Sketches and drawings
2. conceptual CAD design without dimensional constraints
3. adding in components into concept
4. Iterating dimensions to fit calculations and component sizes
5. Optimisation



1. CAD PICS

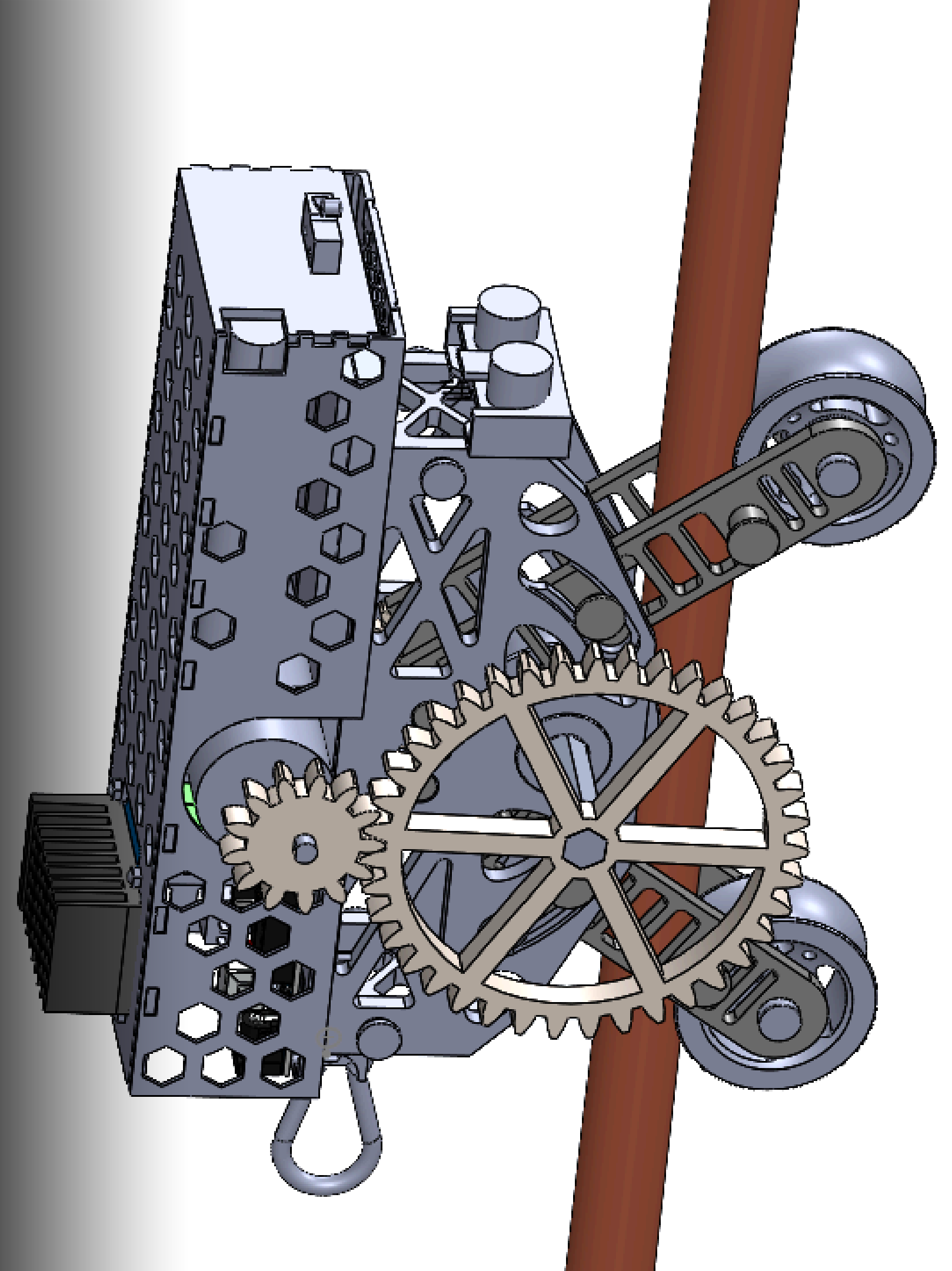
DESIGN ITERATIONS



1.ADD iterations on CAD

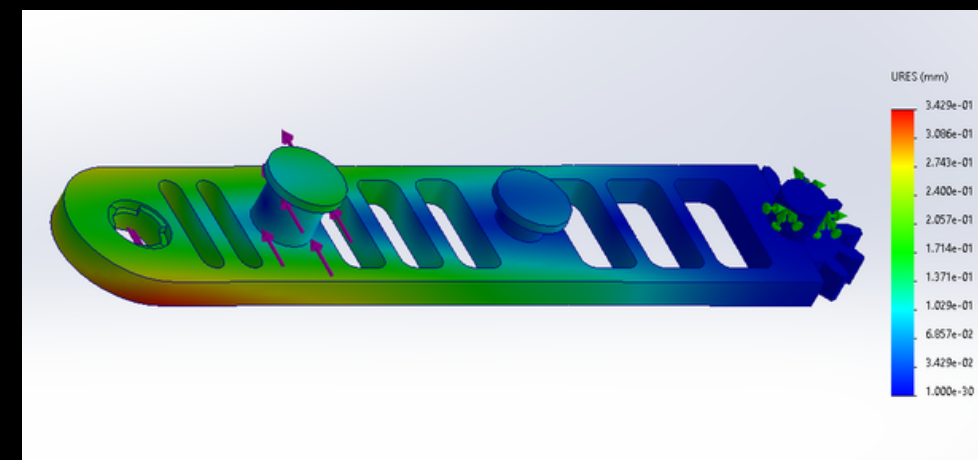
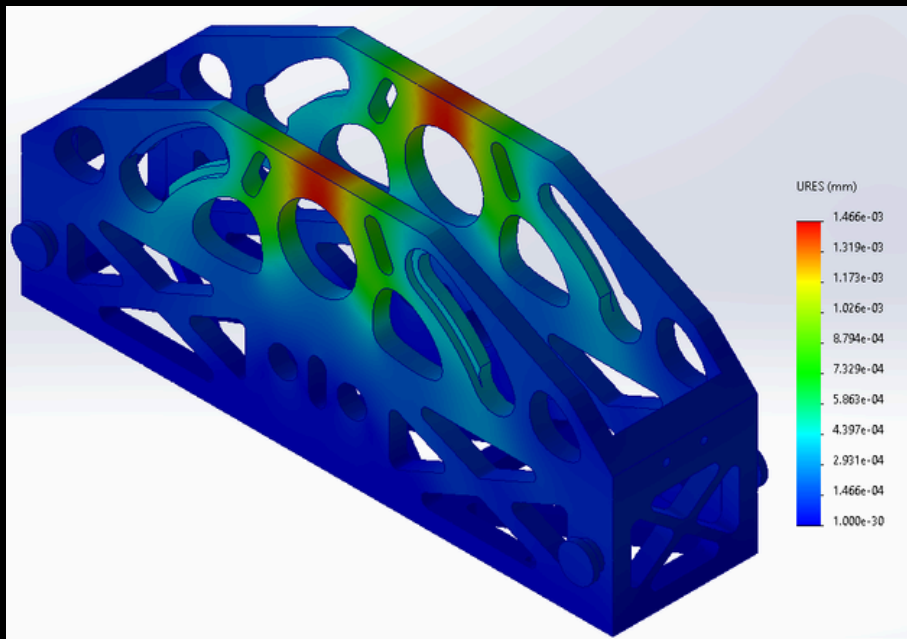
FINAL DESIGN

- Rubber band clamp system
- 3-wheel design
- Middle wheel drive
- moment assisted grip
- rubber wheels
- press-fitted casing
- solenoid release mechanism



OPTIMISATION

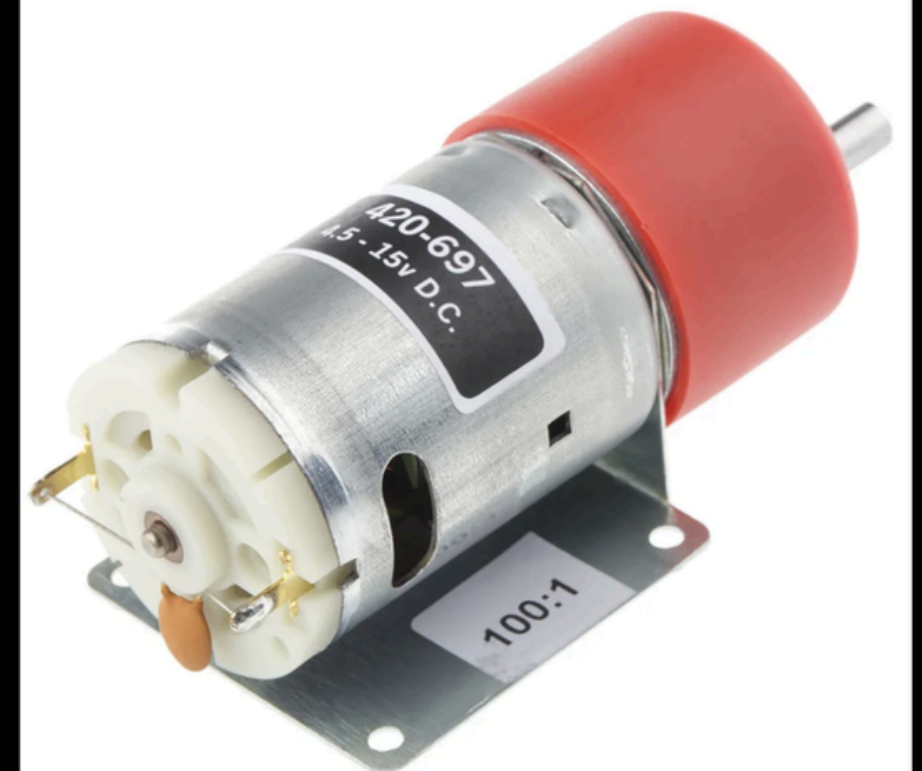
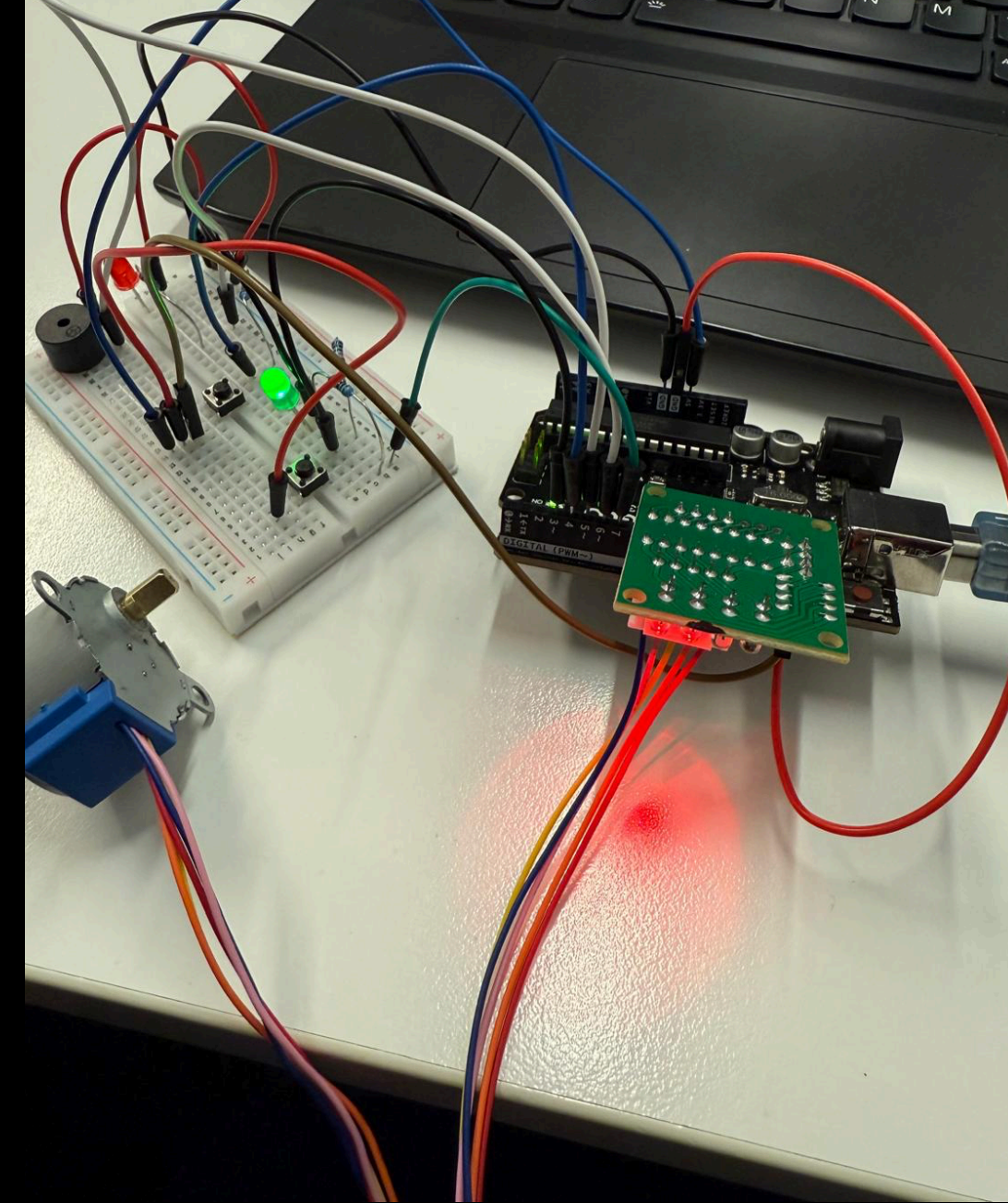
- **Topology optimised parts:**
 - Reduced material usage
 - Removing materials from areas of low stress concentration



COMPONENT SELECTION

Selected Electronics:

1. Arduino Uno R3
2. RS PRO, 12 V dc Brushed DC Geared Motor
3. HC-SR04 ultrasonic sensor
4. 43A H-bridge high-power motor driver module
5. Electromagnetic stroke micro solenoid
6. 12V 2400mAh NiMH rechargeable battery



COSTS

- **Total price: £82.48**

